Flight Delay Prediction

Data Bootcamp Final Project | Helen Tu & Siye Li

5.8M Flight Records	**5** ML Models	**200K** Rows Trained

Can we predict if a flight will arrive 15+ minutes late using only pre-departure information?

The Problem & Dataset

Predictive Task

Binary classification: arrival 15+ min late?

Target:
ARRIVAL_DELAY > 15 min (FAA standard)

Features (pre-departure only):

- Airline, Origin / Destination airport
- Scheduled departure hour, Month, Day of week
- Scheduled flight time, Distance

5.8M Total 2015 USDOT records

200K Sampled for modeling

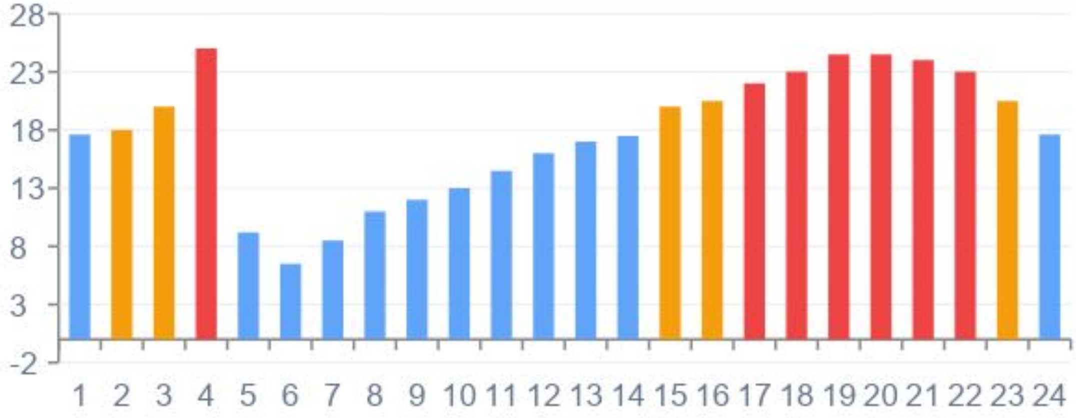
82.4% On-time flights (majority)

17.6% Delayed — our target class

Key Challenge: Class Imbalance — Without correction, all models predict 'On Time' for everything Recall = 0%

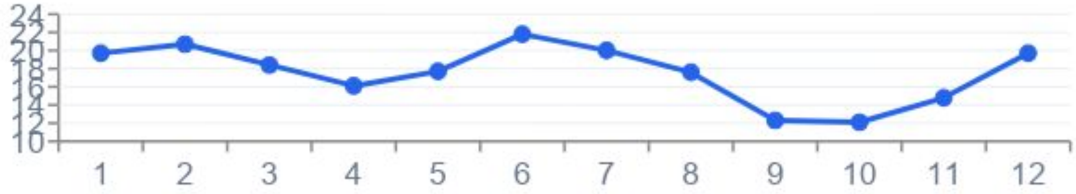
EDA: When Do Delays Happen?

Delay Rate by Hour of Day



Early morning (5-6am): ~6.5% | Evening (18-21h): >24% → cascading delays accumulate

Delay Rate by Month



📅 Worst Month

June

21.8% delay rate

📅 Best Month

October

12.1% delay rate

📅 Worst Day

Thursday

~19% delay rate

📅 Best Day

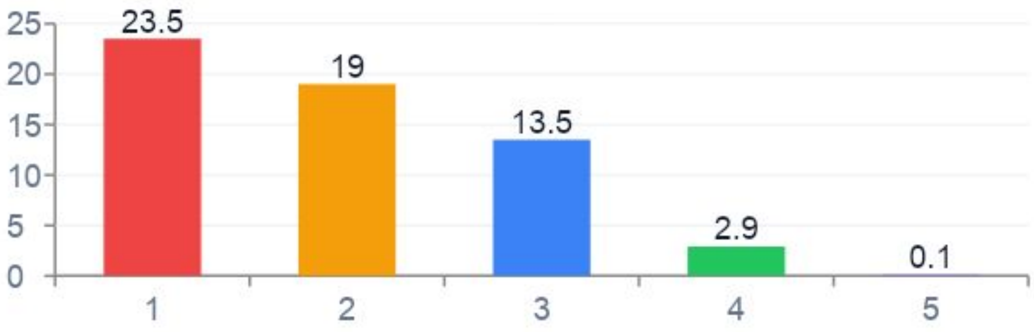
Saturday

~15% delay rate

EDA: Airlines, Airports & Delay Causes

Delay Rate by Airline

Delay Cause — Avg Minutes



Weather = longest delays (23.5 min avg) | Late Aircraft = most frequent cause year-round | Weather peaks: Jan/Feb & Jun/Jul

Top Delayed Airports

CID ~27%

HPN ~25%

XNA ~24%

LGA ~23%

BTR ~23%

MIA ~22%

ORD ~22%

Cancellations (1.5% of flights)

Weather 54.3%

Airline 28.1%

NAS 17.5%

Feb worst month (4.8%) | American Eagle worst airline (5.1%)

Models & Methodology

Logistic Regression	KNN (k=5)	Decision Tree	Naive Bayes	Random Forest
Linear	**Distance**	**Tree**	**Bayes**	**Ensemble**
Imbalance fix:	Imbalance fix:	Imbalance fix:	Imbalance fix:	Imbalance fix:
class_weight ='balanced'	50/50 Downsample	class_weight ='balanced'	Probabilistic Baseline	class_weight ='balanced'

Without class_weight='balanced': Accuracy = 82.4% but Recall = 0% (all predictions = On Time).

Fix: Upweight Delayed class by 4.7x = forces the model to actually learn delay patterns.

sklearn Pipeline:

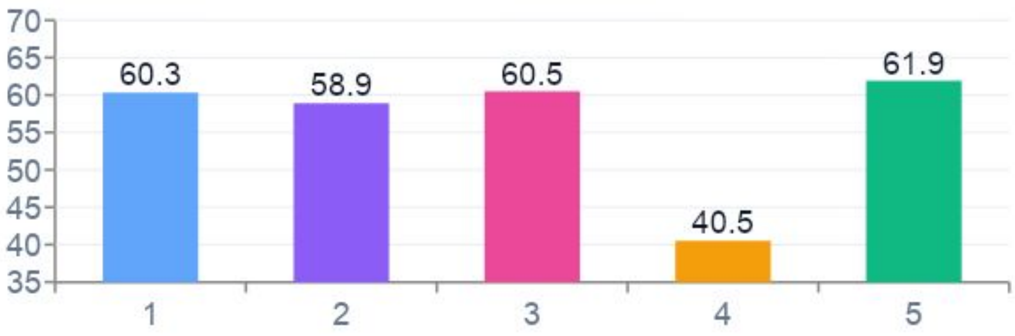
Raw Features -> OneHotEncoder (categoricals) -> StandardScaler (numerics) -> Classifier Model -> Predict 0 / 1

GridSearchCV: Decision Tree (recall-optimized, best depth=1) | Random Forest multi-class (accuracy-optimized, best depth=None)

Binary Classification Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	59.7%	24.2%	60.3%	34.5%	0.633
KNN (k=5)	56.5%	22.2%	58.9%	32.2%	0.597
Decision Tree	60.6%	24.7%	60.5%	35.1%	0.641
Naive Bayes	63.9%	21.8%	40.5%	28.3%	0.576
Random Forest	**60.3%**	**24.8%**	**61.9%**	**35.4%**	**0.653**

Recall Comparison (most important metric for this use case)



For a passenger-facing tool: Recall matters most — better to warn unnecessarily than miss a real delay

Best Recall

RF: 61.9%

Best AUC

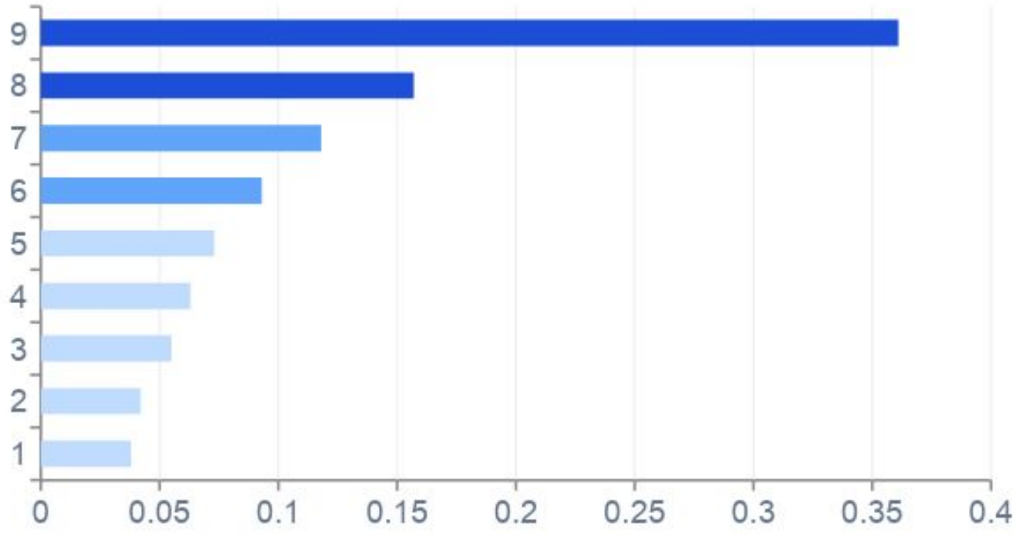
RF: 0.653

Naive Bayes

High acc, low recall

Feature Importance & Lift Analysis

Random Forest — Feature Importance



#1 HOUR

Importance: 0.361

36% of prediction power

Evening > 24% delay rate

#2 AIRLINE

Importance: 0.157

Spirit 28% vs Hawaiian 10%

#3 MONTH

Importance: 0.118

June 21.8% vs October 12.1%

Example Prediction

Southwest ORD to LAX | Friday July 4th | 6:00 PM departure

Predicted delay probability: 61.3% DELAYED

Lift Analysis (Random Forest)

Top 20% highest-risk flights

captures 33.9% of all delays

1.69x lift over random selection

Extended Analysis: GridSearchCV & Profit Curve

GridSearchCV — Decision Tree (recall-optimized)

Parameter	Value
Best max_depth	**1**
CV Recall (5-fold)	**71.4%**
Test Recall	**70.8%**
Test ROC-AUC	0.576

depth=1 tree trades AUC for maximum recall — a single split on HOUR is the most impactful decision

Before vs After GridSearch

Before Recall 60.5%	
TN 20K	**FP 12,975**
FN 2,781	**TP 4,254**

After Recall 70.8%	
TN 14K	**FP 18,327**
FN 2,053	**TP 4,982**

Profit Curve (TP = +$50, FP = -$10)

KNN

Max $1,191,100 @ 59.9% flagged

Decision Tree

Max $418,390 @ 47.0% flagged

Random Forest

Max $413,620 @ 49.9% flagged

Logistic Reg

Max $318,330 @ 53.3% flagged

DT (tuned)

Max $267,100 @ 58.6% flagged

KNN yields highest profit ($1.19M) despite lower AUC — ranking quality at the decision boundary matters

Extended: Multi-Class Delay Prediction

4 Classes

On Time

DELAY <= 15 min | 161,787 (80.9%)

Short Delay

15 to 45 min | 20,133 (10.1%)

Long Delay

45+ min | 15,041 (7.5%)

Cancelled

CANCELLED = 1 | 3,039 (1.5%)

Multi-Class Results

Model	Train Acc	Test Acc
Logistic Regression	42.1%	42.1%
Decision Tree	41.9%	40.6%
Random Forest (tuned)	**99.9%**	**80.6%**

Key Findings

RF dominates

80.6% test accuracy — far ahead of LR (42%) and DT (41%)

Overfitting

Train 99.9% vs Test 80.6% — needs max_depth constraint to regularize

Best params

GridSearchCV: max_depth=None, n_estimators=100, CV acc=80.7%

On Time recall

RF identifies On Time well (99% recall) but struggles with minority delay classes (~1%)

Conclusion & Next Steps

Best Binary Model

Random Forest
Recall 61.9% | AUC 0.653
1.69x lift over random

Top Feature

Departure Hour
0.361 importance
Evening >24% delay rate

Best Business ROI

KNN Profit Curve
$1.19M max profit
@ 59.9% flagged

Limitations

2015 data only — post-COVID patterns differ significantly

No real-time weather features (wind, precipitation, snow)

RF multi-class overfits: train 99.9% vs test 80.6%

Trained on 200K sample from 5.8M total records

Next Steps

Add NOAA weather data — expected biggest performance gain

Regularize RF multi-class: constrain max_depth

Benchmark XGBoost vs Random Forest on both tasks

Use time-based split: train Jan-Sep, test Oct-Dec